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2. Let $\{f_n\}_{n \in \mathbb{N}} \subset L^+(X, \mathcal{M}, \mu)$ such that $f_n \rightarrow f$ pointwise and such that $\lim \int_X f_n d\mu = \int_X f d\mu < \infty$. Let $E \in \mathcal{M}$. Note that $1_E |f - f_n| \leq |f - f_n|$ for all $n \in \mathbb{N}$. Therefore $\lim_{n \rightarrow \infty} \int_E |f - f_n| d\mu \leq \lim_{n \rightarrow \infty} \int_X |f - f_n| d\mu = 0$. Therefore $\lim_{n \rightarrow \infty} \int_E |f - f_n| d\mu = 0$ which implies that $\lim_{n \rightarrow \infty} \int_E f_n d\mu = \int_E f d\mu$.
3. Let μ be the Lebesgue measure on $\mathbb{R}$, $f \in L^1(\mathbb{R}, \mathcal{B}_{\mathbb{R}}, \mu)$ and for $n \in \mathbb{N}$ define $f_n(x) = f(x) \sin^n(x)$. We want to show that $\lim_{n \rightarrow \infty} \int_{\mathbb{R}} f_n d\mu = 0$. We claim that $f'(x) = \lim_{n \rightarrow \infty} |f_n(x)| = \begin{cases} |f(x)| & x = \pi(n + \frac{1}{2}), n \in \mathbb{Z} \\ 0 & \text{otherwise} \end{cases}$. Note if $x \neq \pi(n + \frac{1}{2}), n \in \mathbb{Z}$, we have that $|\sin(x)| < 1$. Therefore $\lim |\sin^n(x)| = 0$. Otherwise, if $x = \pi(n + \frac{1}{2})$ then we have that $|\sin(x)| = 1$. Therefore $\lim |f(x) \sin^n(x)| = |f(x)|$. Additionally since for any $n \in \mathbb{N}$, $|f(x) \sin^n(x)| \leq |f(x)|$, and $|f(x)| \in L^1(\mathbb{R}, \mathcal{B}_{\mathbb{R}}, \mu)$ then by the LDCT we have that $\int_{\mathbb{R}} f'(x) d\mu = \lim_{n \rightarrow \infty} \int_{\mathbb{R}} |f_n(x)| d\mu$. Note that since f' is non-zero on a countable set, then necessarily $\int_{\mathbb{R}} f'(x) d\mu = 0$. Thus $\lim_{n \rightarrow \infty} \int_{\mathbb{R}} f_n(x) d\mu = 0$.
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